Unit: Unit 0: Foundations of Algebra
Topic: GCF LCM

Name: ________________
Date: __________

Advanced (Habanero)

Q1. What is the GCF of 12 and 18?
(A) 1
(B) 2
(C) 3
(D) 6
(E) 12

Q2. What is the LCM of 9 and 12?
(A) 18
(B) 24
(C) 27
(D) 36
(E) 48

Q3. Find the GCF of 24, 30, and 36 using prime factorization.
(A) 1
(B) 2
(C) 3
(D) 6
(E) 12

Q4. A bookshelf has 12 shelves, and each shelf can hold 8 books. What is the LCM of 12 and 8?
(A) 12
(B) 16
(C) 24
(D) 32
(E) 48

Q5. What is the GCF of 48 and 60?
(A) 1
(B) 2
(C) 4
(D) 12
(E) 24

Q6. A group of friends want to share some candy equally. If they have 18 pieces of candy and there are 3 friends, what is the LCM of 18 and 3?
(A) 9
(B) 12
(C) 18
(D) 27
(E) 36

Q7. Find the GCF of 20, 28, and 32 using factors.
(A) 1
(B) 2
(C) 4
(D) 8
(E) 12

Q8. What is the LCM of 10 and 15?
(A) 15
(B) 20
(C) 25
(D) 30
(E) 45

Open Ended Questions (FRQ)

Q9. Tom has 18 boxes of pens, and each box contains 12 pens. If he wants to share the pens equally among 6 friends, what is the LCM of 18 and 6, and how many pens will each friend get?

Q10. A bakery sells 24 loaves of bread per day, and each loaf is cut into 8 slices. What is the GCF of 24 and 8, and how many slices of bread are sold per day?

Chef's Tips

- Tip 1: Emphasize the importance of prime factorization for finding GCF and LCM.
- Tip 2: Use real-world examples to illustrate the applications of GCF and LCM.
- Tip 3: Encourage students to check their work and use multiple methods to verify their answers.